TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Identification of Miscellaneous Organic Compounds

The classes left over from Experiments 1–3 — phenol, aniline, the amides, nitrobenzene and glucose — each caught by its own signature colour reaction.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 4 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to identify an unknown drawn from the remaining common classes: a **phenol** (phenol itself), a primary aromatic **amine** (aniline), an **amide** (urea or acetamide), a **nitro compound** (nitrobenzene) and a **carbohydrate** (glucose, with sucrose as its non-reducing partner).

There is no single group test for a miscellaneous set — instead, each class carries a chemistry the others cannot imitate. **Solubility behaviour is the first sieve:** phenol dissolves in NaOH (it is acidic) but gives no CO_2 with $NaHCO_3$ (Experiment 2 — it is weaker than carbonic acid); aniline, a base, dissolves in dilute HCl and is thrown out again by NaOH; urea and glucose dissolve freely in water to neutral solutions; nitrobenzene, a heavy oil, dissolves in neither. Then come the signatures. Phenol's activated ring couples and brominates instantly (white tribromophenol) and its ferric chloride colour is violet. Aniline shares the instant bromination but its royal road is **diazotisation**: at 0–5 °C nitrous acid converts $–NH_2$ into a diazonium salt that couples with β-naphthol to a blazing orange-red azo dye — no other class on this list can do it. Amides hydrolyse in hot alkali and breathe out ammonia. A nitro group is identified by reducing it to something already known: reduce nitrobenzene and the product answers aniline's tests. Glucose declares itself as a reducing sugar, and the Molisch test vouches for carbohydrates as a whole.

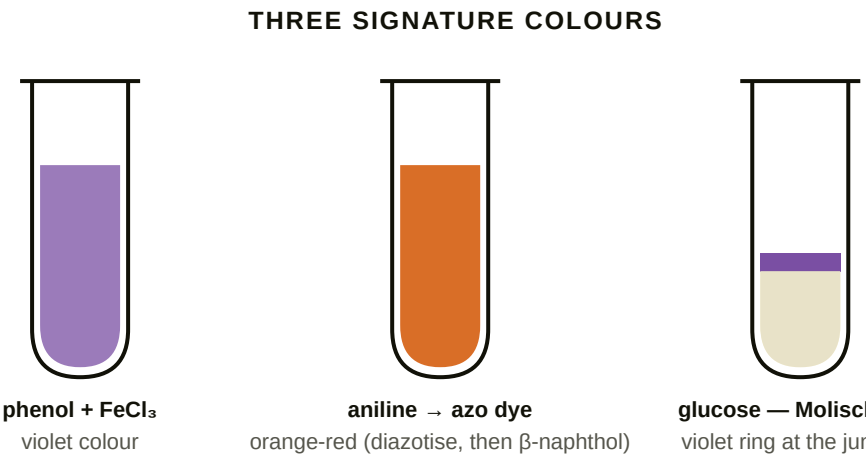


Figure 1 — Three of the signature reactions of this experiment. Colours are identifications only when the rest of the pattern agrees.

2. Apparatus and reagents

- Test tubes, ice bath (for diazotisation), water bath, droppers, red and blue litmus papers, glass rod with HCl for testing ammonia fumes; the unknown compound.
- Dilute HCl, NaOH solution, $NaHCO_3$ solution, neutral $FeCl_3$ solution, bromine water, phthalic anhydride and conc. H_2SO_4 (phthalein test), $NaNO_2$ solution and β-naphthol dissolved in NaOH (diazotisation–coupling), chloroform and alcoholic KOH (carbylamine test), bleaching powder solution, Zn dust with NH_4Cl solution and with conc. HCl (reductions), Fehling's solutions, Tollens' reagent (fresh), Molisch reagent (α-naphthol in ethanol) and conc. H_2SO_4, dilute $CuSO_4$ solution (biuret test).

3. Test scheme

Phenol

- **Neutral $FeCl_3$:** violet (purple) colouration — the phenolic –OH complexing iron.
- **Bromine water:** decolourised at once with a white precipitate of 2,4,6-tribromophenol: $C_6H_5OH + 3\ Br_2 \rightarrow C_6H_2Br_3OH\downarrow + 3\ HBr$. The ring is so activated that all three free ortho/para positions brominate without a catalyst.
- **Phthalein test:** fuse a little phenol with phthalic anhydride and a drop of conc. H_2SO_4; cool, dilute, and make alkaline with NaOH — a pink colour (phenolphthalein itself has been synthesised in the tube).
- **Liebermann's test:** $NaNO_2$ and conc. H_2SO_4 give a deep blue-green colour; on dilution it turns red, and excess NaOH restores the blue (indophenol chemistry).

- Behaviour check: dissolves in NaOH, **no effervescence with $NaHCO_3$** — the line between phenol and a carboxylic acid (Experiment 2).

Aniline (primary aromatic amine)

- **Diazotisation and coupling:** dissolve in dilute HCl, cool to 0–5 °C in ice, add $NaNO_2$ solution — $C_6H_5NH_2 + NaNO_2 + 2\ HCl \rightarrow [C_6H_5N_2]^+Cl^- + NaCl + 2\ H_2O$ — then pour into alkaline β-naphthol: a bright **orange-red azo dye** separates. This is the definitive test for a primary aromatic amine.
- **Bromine water:** decolourised with a white precipitate of 2,4,6-tribromoaniline (like phenol — so bromination alone cannot separate these two; $FeCl_3$ and the diazo test can).
- **Carbylamine test:** warm a drop with chloroform and alcoholic KOH — the unbearably foul odour of phenyl isocyanide: $C_6H_5NH_2 + CHCl_3 + 3\ KOH \rightarrow C_6H_5NC + 3\ KCl + 3\ H_2O$. Primary amines only (aliphatic or aromatic).
- **Bleaching powder solution:** a drop gives a purple-violet colour (oxidation products of aniline).
- Behaviour check: insoluble in water; dissolves in dilute HCl (basic); reprecipitated by NaOH.

Amides — urea and acetamide

- **Alkaline hydrolysis:** boil with NaOH solution — ammonia is evolved (turns moist red litmus blue; dense white fumes with a rod dipped in conc. HCl): $CH_3CONH_2 + NaOH \rightarrow CH_3COONa + NH_3\uparrow$; for urea, $CO(NH_2)_2 + 2\ NaOH \rightarrow Na_2CO_3 + 2\ NH_3\uparrow$.
- **Biuret test (urea):** heat solid urea gently until it melts and ammonia stops coming off (two molecules have condensed to biuret, $H_2N–CO–NH–CO–NH_2$); cool, dissolve in water, add NaOH and one drop of very dilute $CuSO_4$ — a **violet** colour. The colour needs the –CO–NH–CO– grouping, the same linkage as in proteins.
- **Hypobromite test (urea):** sodium hypobromite solution gives brisk effervescence of nitrogen: $CO(NH_2)_2 + 3\ NaOBr \rightarrow N_2\uparrow + CO_2 + 3\ NaBr + 2\ H_2O$ (in alkali the CO_2 is held as carbonate). The basis of a classical estimation of urea.

Nitrobenzene

- Yellowish, heavy, oily liquid with a bitter-almond odour (easily confused with benzaldehyde — but it gives **no** 2,4-DNP precipitate and no Tollens mirror).
- **Reduction–identification:** warm with Zn dust and NH_4Cl solution; the product, phenylhydroxylamine, reduces Tollens' reagent (Mulliken–Barker test) — the nitro group has been half-reduced into a reducing agent. Or reduce fully with Sn/conc. HCl to aniline and prove the aniline by diazotisation–coupling (§3 above): the unknown is identified through its reduction product.

Glucose (and sucrose)

- **Molisch test (all carbohydrates):** add 2 drops of α-naphthol solution to the aqueous unknown, then pour conc. H_2SO_4 carefully down the side of the tilted tube — a **violet ring** at the junction of the two layers. The acid dehydrates the sugar to furfural derivatives, which condense with α-naphthol to the violet dye.
- **Fehling's solution, boil:** red Cu_2O precipitate — glucose is a reducing sugar (free aldehyde group in its open-chain form). Tollens' reagent likewise gives a mirror.
- **Barfoed's test:** the acidic copper acetate reagent is reduced within about 1–2 minutes of boiling by a monosaccharide (glucose); a disaccharide such as sucrose reacts only after much longer heating — the timing separates them.
- **Sucrose:** Molisch positive but Fehling **negative** — until the solution is boiled with a few drops of dilute HCl, neutralised with NaOH, and retested: hydrolysis has freed glucose and fructose, and Fehling now reduces briskly.

4. Summary decision table

COMPOUND	SOLUBILITY CLUE	KEY TEST	OBSERVATION
Phenol	Dissolves in NaOH; no CO_2 with $NaHCO_3$	Neutral $FeCl_3$; bromine water	Violet colour; white ppt of 2,4,6-tribromophenol
Aniline	Dissolves in dilute HCl; thrown out by NaOH	Diazotise (0–5 °C) + alkaline β-naphthol	Bright orange-red azo dye
Urea	Very soluble in water; neutral solution	Biuret test; warm NaOH	Violet colour; NH_3 evolved on boiling with alkali
Acetamide	Soluble in water; neutral solution	Boil with NaOH	NH_3 evolved (no biuret — that needs urea's paired groups)
Nitrobenzene	Heavy oil; insoluble in water, acid and alkali	Reduce (Zn/NH_4Cl), then Tollens; or Sn/HCl then diazo test	Reduction product reduces Tollens / couples as aniline
Glucose	Soluble in water; neutral solution	Molisch; Fehling (boil)	Violet ring; red Cu_2O at once
Sucrose	Soluble in water; neutral solution	Molisch; Fehling before/after hydrolysis	Violet ring; Fehling negative until hydrolysed, then positive

5. Model identification record

WORKED EXAMPLE — UNKNOWN D

White crystalline solid, very soluble in water, solution neutral to litmus.
Fehling (boiled): no reaction. Molisch: no violet ring. → Not a carbohydrate; $NaHCO_3$: silent. → Not an acid. $FeCl_3$: no violet. → Not phenol.
Boiled with NaOH solution: a gas evolves that turns moist red litmus blue and gives white fumes with an HCl rod. → **Ammonia** — an amide is indicated.
A fresh portion is heated dry until it just melts and the ammonia smell fades; the residue, dissolved and treated with NaOH and one drop of dilute $CuSO_4$, gives a **violet** colour (biuret test).
Conclusion: unknown D is urea.

6. Precautions

- **Diazotisation must stay at 0–5 °C.** Benzenediazonium chloride decomposes above about 5–10 °C, losing N_2 and giving phenol — a warm tube couples weakly or not at all, and a good unknown gets misreported.
- Carbylamine test: the isocyanide product is poisonous and its odour overwhelming — a single drop of amine, fume cupboard, and waft once only. Destroy the mixture with concentrated HCl afterwards.
- Bromine water is corrosive and its vapour attacks eyes and lungs — dispense in the fume cupboard, gloves on, and never pipette by mouth.
- In the Molisch test the acid layer must be poured slowly down the tilted tube's side: mixing destroys the ring and the violent heat of mixing can spit hot acid.
- Phenol burns skin (white patches, anaesthetic — you may not feel it) and aniline is absorbed through skin: wash any contact at once and report it. Never taste an unknown to check a "sweet" carbohydrate.
- Tollens' residues are destroyed with dilute HNO_3 immediately (Experiment 3).

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Both phenol and a carboxylic acid dissolve in NaOH. Which single test separates them, and why does it work?
2. Why is diazotisation carried out in an ice bath, and what goes wrong chemically if the mixture warms up?
3. Phenol and aniline both decolourise bromine water with a white precipitate. Give two tests that separate them.

4. For which amines is the carbylamine test valid, and what is the product responsible for the odour?
5. An unknown gives a positive Molisch test but does not reduce Fehling's solution. After boiling with dilute HCl and neutralising, it reduces Fehling's briskly. Name it and explain both results.
6. What structural feature must a compound possess to answer the biuret test, and how does urea — a single small molecule — come to possess it in the test?

Answer key

1. Sodium bicarbonate solution. A carboxylic acid is stronger than carbonic acid and liberates CO_2 briskly; phenol is weaker than carbonic acid, so no gas is evolved even though both dissolve in the stronger alkali NaOH.
2. The diazonium salt is stable only in the cold. Warmed, it decomposes: $[C_6H_5N_2]^+ + H_2O \rightarrow C_6H_5OH + N_2\uparrow + H^+$, so the coupling partner is lost and the dye test fails or fades.
3. (i) Neutral $FeCl_3$: phenol gives a violet colour, aniline gives none. (ii) Diazotisation–coupling: only aniline (a primary aromatic amine) gives the orange-red azo dye. (Solubility separates them too: aniline dissolves in dilute HCl, phenol does not.)
4. Primary amines only, aliphatic and aromatic. The product is an isocyanide (carbylamine), R–N≡C — phenyl isocyanide from aniline — whose penetrating foul odour is the signal.
5. Sucrose. Molisch is positive because it is a carbohydrate. Its glucose and fructose units are joined through both anomeric carbons, so no free reducing group exists and Fehling stays blue. Acid hydrolysis splits the disaccharide into free glucose and fructose, both reducing sugars — hence the brisk red Cu_2O afterwards.
6. Two –CO–NH– units linked together (the peptide-like grouping, as in biuret and in proteins). Urea alone has only one such unit, but heating two urea molecules condenses them — NH_3 is expelled and biuret, $H_2N–CO–NH–CO–NH_2$, with the paired grouping is formed, which then complexes Cu^{2+} in alkali to give the violet colour.
